



Week 1 Lecture 7

▼ Class	BSCCS2003
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Simple Web Server

Simplest server

```
#!/bin/bash

while true; do
  echo -e "HTTP/1.1 200 OK\n\n $(date)" |
  nc -l localhost 1500;
done
```

Create a new file on the linux terminal

```
$ nano server.sh
```

Write the above mentioned server bash script in that file, and then to run the script, type the following code on the linux terminal

```
$ bash server.sh
```

We can send a request to the running server from another instance of linux shell

```
$ curl http://localhost:1500
```

We can also add a verbose flag to get a ton more info

```
$ curl -v http://localhost:1500
```

What the general web server is actually doing?

- Listening on a fixed port
- On incoming requests, it ran some code, generated results on the fly and returned the results to the client
 - Standard headers to be sent as a part of the result
 - Output can be in text or other format - MIME (Multipurpose Internet Mail Extensions)

What does a typical request look like?

```
GET / HTTP/1.1
Host: localhost:1500
User-Agent: curl/7.64.1
Accept: */*
```

Viewing responses with curl

- curl, wget, etc - simple command line utilities
- Can perform full HTTP requests
- Verbose output includes all the headers
- Very useful for debugging!

What does a typical response look like?

When you make a simple http request to a web server via curl, the following response appears ...

```
*   Trying ::1...
* TCP_NODELAY set
* Connected to localhost (::1) port 1500 (#0)
> GET / HTTP/1.1
> Host: localhost:1500
> User-Agent: curl/7.64.1
> Accept: */*
>
< HTTP/1.1 200 OK
* no chunk, no close, no size. Assume close to signal end
<
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```

NOTE: The `:: 1` is an IPv6 address which represents *localhost*